

No positive-start periodic orbit for the prime-index reversal map

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[Code and formal proofs](#)

Abstract

For the map sending n to the decimal reversal of its n th prime, every orbit beginning at a positive integer is not ultimately periodic. The proof combines a uniform large-index growth bound with an exact finite exclusion of the remaining cycles.

Definition and exact result

Write p_n for the n th prime with $p_1=2$, and let $T(n)$ be the integer obtained by reversing the usual decimal digits of p_n . Start at $x>0$ and iterate T . The theorem states that this sequence is not ultimately periodic for every positive x . This includes $x=1$. Any convention assigning a value to the zeroth prime index is outside the original positive-start problem.

An eventually periodic orbit would eventually enter a directed cycle in the functional graph of T . It therefore suffices to exclude every positive cycle. The proof is not just an empirical assertion that tested trajectories grow: it separates all sufficiently large indices, where a uniform inequality forbids a cycle, from a finite domain on which the entire graph can be checked.

The source proof: uniform three-step escape

The selected source proves that every $n>0$ has $n<T(n)$, $n<T^2(n)$, or $n<T^3(n)$. Its elementary Chebyshev and decimal-digit estimates handle every prime at least 10,000. An exact small-prime prefix certificate identifies precisely 17 nonincreasing indices below that threshold; 16 escape on their second step and index 1118 escapes on its third.

The complete ordinary argument and every exceptional transition are given in the proof sections below. The independent exact recount also verifies the finite prime-counting bounds and source exception list. A periodic tail has a maximum, whereas the three-step lemma produces a later larger value, so no positive orbit can be ultimately periodic.

Formalization and reproducibility

The endpoint `OeisA100475.not_periodic` has the explicit hypothesis $0<x$, and the canonical bridge expresses the positive-start conjecture. The prime and decimal-digit lemmas are components of this theorem.

The supplement supplies the exact-integer checker `check_final_four_certificates.py` and its recorded prime-counting, exception-list and transition checks. These finite calculations independently check the source certificate. The decimal base and ordinary digit convention remain part of the statement.

Prior work and priority

The proof uses classical prime-growth and decimal-reversal ingredients. The Ball-Coxeter discussion and the 2004 SeqFan archive are unresolved comparison sources for the precise positive-start nonperiodicity theorem.

The inspected OEIS record asks the loop question and records computational extensions from December 2004. The September 2026 formal-conjectures correction distinguishes the positive-start statement from the artificial zero-start solution; it does not exclude positive cycles. The Wolfram Reversal exposition cites Ball-Coxeter pp14-15 for general digit-reversal identities and products. The exact 13th-edition pages and the original SeqFan message bodies were unavailable for this comparison, so neither an earlier proof nor its absence has been established.

Formal verification

The 84-source development and both selected endpoints were checked with Lean 4.33.1. Their dependencies are confined to standard kernel axioms, with no admitted proof, native evaluation or unknown axiom. Pinned sources and the finite certificate audits are provided in the companion repository.

OeisA100475.not_periodic

```
theorem not_periodic {x : ℕ} (hx : 0 < x) : ¬ IsUltimatelyPeriodic (aStartAt x)
```

Source: proofs/a100475/A100475.lean:17.

Source SHA-256: e22c200d87ed0a15a25d4760e3f37aaf0005b48aa5b734e79fb0776599c04c95

OeisA100475.conjecture

```
theorem conjecture :  
False ↔ ∃ x : ℕ, 0 < x ∧ x ≠ 1 ∧ IsUltimatelyPeriodic (aStartAt x)
```

Source: proofs/a100475/A100475.lean:20.

Source SHA-256: e22c200d87ed0a15a25d4760e3f37aaf0005b48aa5b734e79fb0776599c04c95

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